

FX SENSOR INTERFACE REGISTER MAP		
FX devices	Meaning	Access / validation
D200..207	Staging raw8	CIMON write
D208	Prepare ID	CIMON write; new > high-water
D209	Commit ID	CIMON write; prepared ID
D210	Reset request ID	CIMON write; new reset transaction
D220..227	Prepared fixed snapshot	FX owned
D228	Prepared ID	FX read
D230..237	Applied fixed snapshot	FX read
D238	Applied ACK ID	FX read
D239	Reset ACK ID	FX read
D240..255	Preview8 DWORDs	FX read; D240,D242,...D254
D257	Highest accepted prepare ID	FX read; reset request does not clear

PREVIEW FORMAT

D240 low-word=prepared MAP; D241 high-word=prepared ID.
Next pair D242/D243 = BOOT + prepared ID.
All eight OPC items: unsigned32, adjacent2 D words.
Do not register a textual D240~D241 range.

TRANSACTION BOUNDARY

Only equal prepared values can be committed.
A new COMMIT applies the validated prepared snapshot.
D230..237 keeps the fixed accepted snapshot.
Active D7 ages between commits in 100ms steps.
When M50=0: D5=0, D6=1 and D7=32000.
Same-session transaction IDs1..32000 not reused.
Reset clears prepared/ACK state, not D257 high-water.

normal_v01 uses4s transaction limit and same-ID prepare retries. FX timing T0=10s /T1=6s /AGE<12s is user-confirmed2026-10-13.

No direct ESP→FX serial link; CIMON writes FX OPC staging items and waits for read-back evidence.

NO.	DATE	REVISION
R0	2026-10-22	FUNCTION RECORD

DWG. TITLE

SEMICON SORTER PROJECT
Final Test and Multi-Bin Sorting Cell

FX SENSOR PREPARE / COMMIT / ACK

DRAWN BY	DATE
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DWG. NO
011

SHEET NO
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